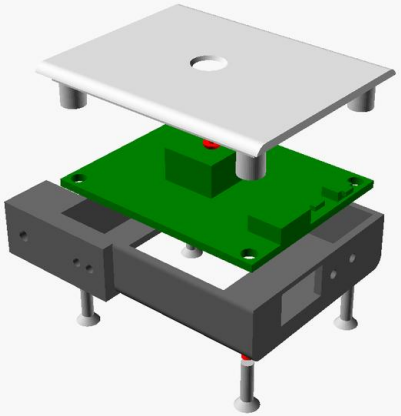
tCam-Mini mount for the Bambu Lab A1: assembly and installation

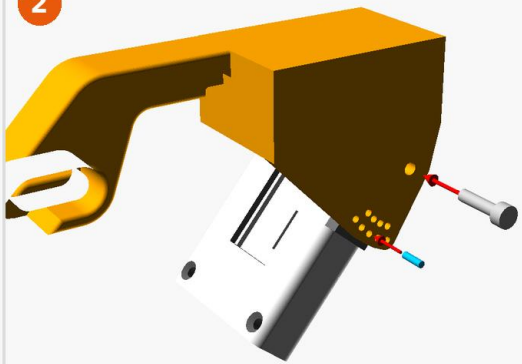
1



Close up the case

Board in case_base_a1: USB-C in the end opening, ESP32 can into the floor cut-out. Lid on (Lepton in its recess), 4 x M2.5x10 countersunk screws from the back. Lug end = top.

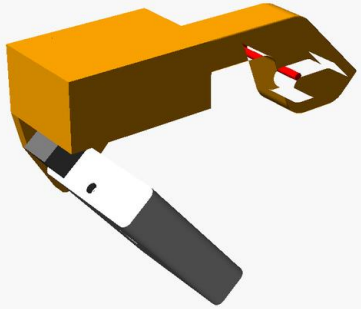
2



Set the angle (off the printer)

Push the 6 mm filament pin through the plate hole for your angle into case hole A or B (chart, step 6). Then the M3x12 pivot screw through the plate into the lug, snug only.

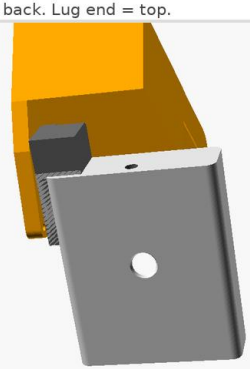
3



Slide onto the cutter trigger

Take the old mount off the trigger at the right end of the gantry. Slide the bracket's socket onto the trigger the same way (arrow), all the way home. Camera looks left along the gantry.

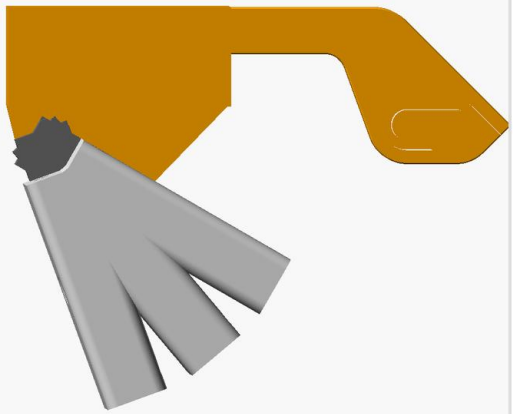
4



Route the cable

USB-C leaves the bottom end of the case. Leave a loop: the camera rides up and down with the gantry. Keep the cable clear of the toolhead, the PTFE tube and the cutter lever.

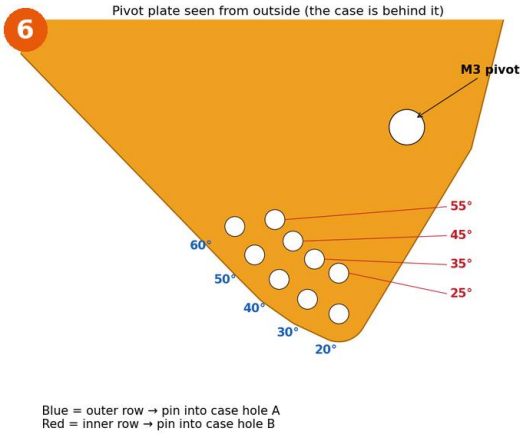
5



Check by hand before printing

Motors off. Push the toolhead along the whole gantry and right up against the trigger. Watch the case and cable. Ghosts: the case at 20 and 60 deg; it swings toward the trigger as it tilts.

6



Angle holes

Start at 40 deg. 20 deg sees furthest along the gantry; 60 deg looks more steeply down near the right end. To change it, remove the pin, tilt the case, re-insert the pin.

Hardware

Fasteners

1 x M3x12 screw: pivot, self-taps into the lug
4 x M2.5x10 countersunk screws: close the case
6 mm of 1.75 mm filament: angle-lock pin

Printed parts

bracket_a1.stl: on its back, supports from the build plate under the pivot plate only
case_base_a1.stl + case_lid.stl: no supports
PLA or PETG (open printer, no heat issue)

Cable and credits

USB-C cable with a loop for Z travel
Trigger socket: grizzie17's Wyze Cam3 A1 mount (CC BY-NC-SA 4.0), so bracket_a1.stl is too